eProject Guide - Building Applications on Java EE Platform

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First Edition - 2015





Dear Learner,

We congratulate you on your decision to pursue an Aptech Worldwide course.

Aptech Ltd. designs its courses using a sound instructional design model – from conceptualization to execution-incorporating the following key aspects: –

- Scanning the user system and needs assessment

Needs assessment is carried out to find the educational and training needs of the learner.

Technology trends are regularly scanned and tracked by core teams at Aptech Ltd. TAG* analyzes these on a monthly basis to understand the emerging technology training needs for the Industry.

An annual Industry Recruitment Profile Survey# is conducted during August - October to understand the technologies that Industries would be adapting in the next 2 to 3 years. An analysis of these trends and recruitment needs is then carried out to understand the skill requirements for different roles and career opportunities.

The skill requirements are then mapped with the learner profile (user system) to derive the learning objectives for the different roles.

- Needs analysis and design of curriculum

The learning objectives are then analyzed and translated into learning tasks. Each learning task or activity is analyzed in terms of knowledge, skills and attitudes that are required to perform that task. Teachers and domain experts do this jointly. These are then grouped in clusters to form the subjects to be covered by the curriculum.

In addition, the society, the teachers, and the industry expect certain knowledge and skills that are related to abilities such as learning-to-learn, thinking, adaptability, problem solving, positive attitude etc. These competencies would cover both cognitive and affective domains.

A precedence diagram for the subjects is drawn, where the prerequisites for each subject are graphically illustrated. The number of levels in this diagram is determined by the duration of the course in terms of number of semesters etc. Using the precedence diagram and the time duration for each subject, the curriculum is organized.

- Design and development of instructional materials

The content outlines are developed by including additional topics that are required for the completion of the domain and for the logical development of the competencies identified. Evaluation strategy and scheme is developed for the subject. The topics are arranged/organized in a meaningful sequence.

The detailed instructional material – Training aids, Learner material, reference material, project guidelines, etc. are then developed. Rigorous quality checks are conducted at every stage.

- Strategies for delivery of instruction

Careful consideration is given for the integral development of abilities like thinking, problem solving, learning-to-learn etc. by selecting appropriate instructional strategies (training methodology), instructional activities and instructional materials.

The area of IT is fast changing and nebulous. Hence considerable flexibility is provided in the instructional process by specially including creative activities with group interaction between the students and the trainer. The positive aspects of web based learning –acquiring information, organizing information and acting on the basis of insufficient information are some of the aspects which are incorporated in the instructional process.

- Assessment of learning

The learning is assessed through different modes – tests, assignments and projects. The assessment system is designed to evaluate the level of knowledge and skills as defined by the learning objectives.

- Evaluation of instructional process and instructional materials

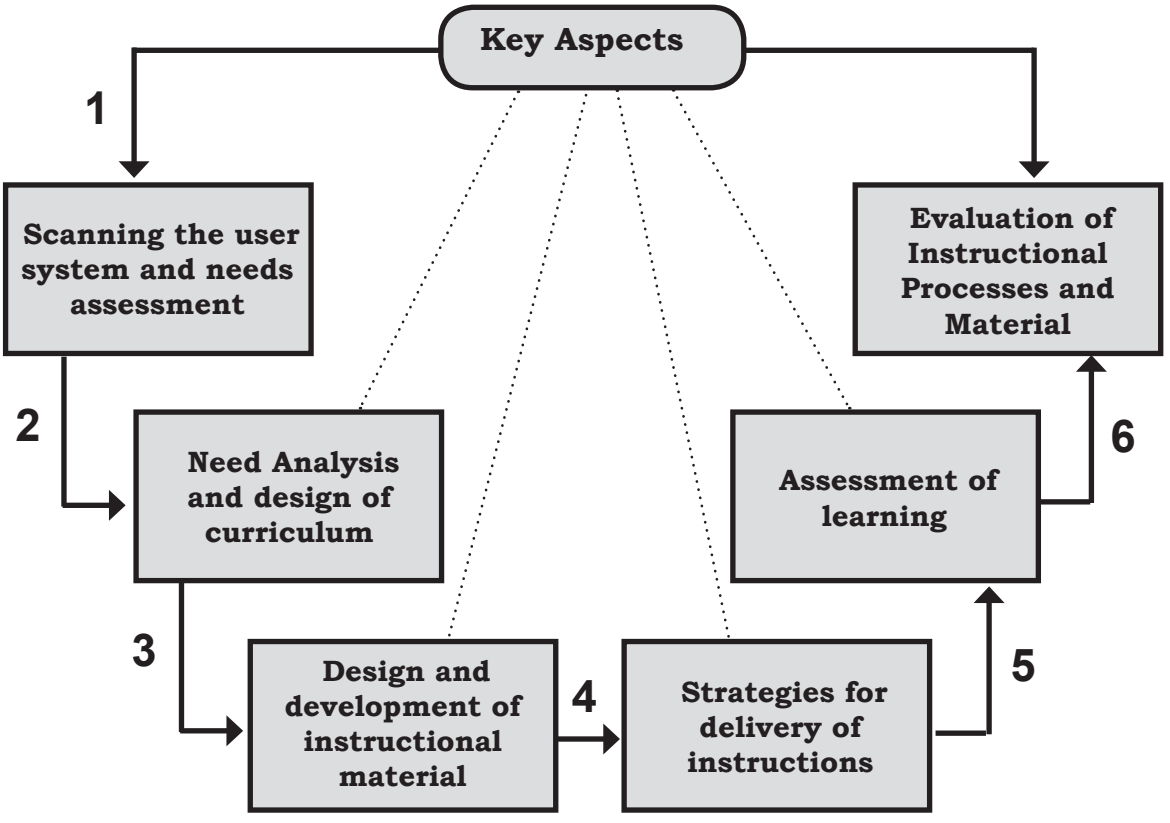
The instructional process is backed by an elaborate monitoring system to evaluate - on-time delivery, understanding of a subject module, ability of the instructor to impart learning. As an integral part of this process, we request you to kindly send us your feedback in the reply pre-paid form appended at the end of each module.

*TAG – Technology and Academics Group comprises members from Aptech Ltd., professors from reputed academic institutions, senior managers from industry, technical gurus from software majors and representatives from regulatory organizations/forums.

Technology heads of Aptech Ltd. meet on a monthly basis to share and evaluate the technology trends. The group interfaces with the representatives of the TAG thrice a year to review and validate the technology and academic directions and endeavors of Aptech Ltd.

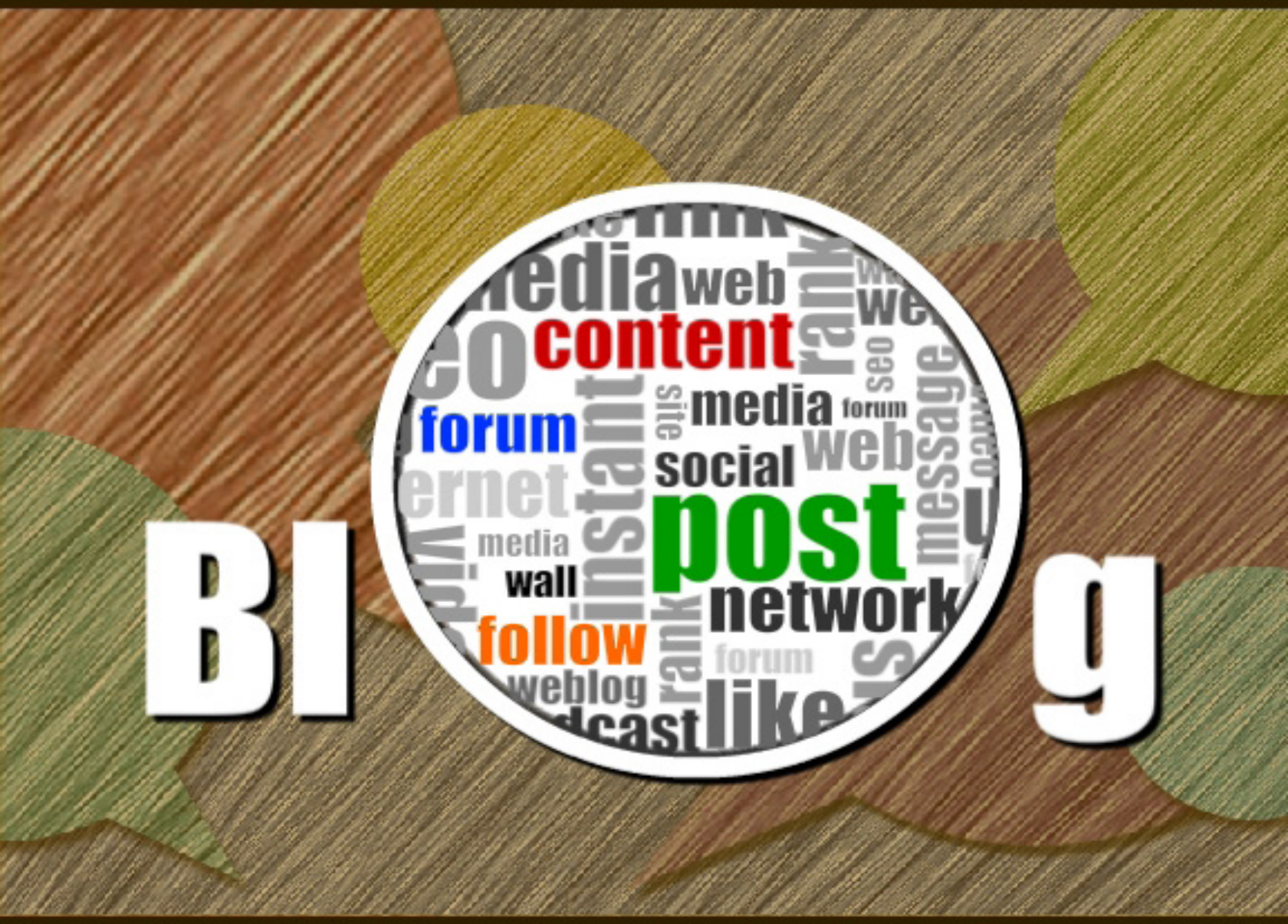
#Industry Recruitment Profile Survey - The Industry Recruitment Profile Survey was conducted across 1581 companies in August/September 2000, representing the software, manufacturing, process industry, insurance, finance and service sectors.

Aptech New Products Design Model



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Preface

The eProject guide provides all the necessary information to students in developing a solution for the work process. Important functionalities required in the application are explained. Students have to understand the workflow and design the project accordingly.

The project specification is organized in the following order:

Introduction provides information related to the software industry. Process flow, suggested modules of the application and their description are also dealt with in the introduction.

Documentation describes different formats for collecting and maintaining the information. These can be used as guidelines.

Case Study provides source of information that can be used by the students while developing the project.

Sample eProject is for understanding the different aspects covered in project and the way of presenting them.

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Introduction to the Book

The best way to learn something is to apply its principles and then to test them. Similarly, the best way to evaluate the knowledge of the students is to test its application through project work. The degree of success of the project depends on the quality of the Guide. Hence, this Project Guide, which has been prepared following the best practices in the industry, will help the students to have the experience of going through a live project and will teach them the essentials of successful development of IT projects.

The Project Guide will help you to:

- Analyze a project
- Design the specifications of the project
- Develop the solution
- Maintain disciplined documentation for the work done
- Work in groups

The ability to work in a group is a very vital quality for anybody desiring to join the software industry. Your project group will consists of 3-4 members. The Faculty will assign you to a project group and select the Leader for the Group (Group Leader).

This Project Guide reiterates the commitment of eProjects Team in keeping up its tradition of providing innovative, career oriented professional education.

Religiously following the given systematic approaches in this e-project guide would prepare the students to get the real life experience of handling projects, as the practices listed here have been extracted from the current industry norms. Thus, such an exercise would prepare you for joining the software development industry.

Wishing you the very best.

eProjects Team, HO Mumbai.

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Building Applications on Java EE Platform

1. Introduction

1.1 Team

Implementation of any new computerized system involves a team of people. An ideal team consists of the following members:

- Project Manager: The Project Manager coordinates the project. Apart from being knowledgeable about software, the person chosen as the Project Manager must have good writing skills. He must have enough experience in this field so that he can be successful in the implementation. He is involved not only in team management but also in resource allocation, project planning, reporting and such activities; all of which form a part of his responsibilities.

- Project Leader: The Project Leader leads the project team. A Project Leader essentially decides which tasks are to be performed by each team member and how much time should be allotted to each Project Development Phase.

- Analyst: The Analyst studies the requirements of the system and defines the problem. The Analyst determines the inputs, outputs and processes involved in transforming those inputs into outputs. The Analyst acts as the “Technical Expert” and normally studies the technologies that are to be implemented to develop the project.

- Developer/Programmer: The Developer builds the user interface according to the specifications prepared by the Analyst. Next, the Developer builds a prototype of the system. After receiving client approval on the prototype, the Developer adds the necessary code to make the prototype a full-fledged system.

- Tester: The Tester tests the functionality of the application. Test data is used to check if the program is able to execute without causing any errors. Test data may be live data extracted from existing records in the system, or dummy data. The Tester then also verifies the integrated application’s functionality with test data.

- Implementation Engineer: The Implementation Engineer ports the final product to the Client’s computers. The Implementation Engineer will ensure that the installation process has been carried out accurately, and hands over the system to the client.

- Maintenance Engineer: The Maintenance Engineer is responsible for taking care of maintaining the system that has been built. Maintenance includes extending troubleshooting support, and performing software upgrades in case of changes in the external system.

1.2 Tips for a Good Application

The seven basic steps for creating an effective Web-based application are:

1. Define your target audience
2. Organize your concepts, information and material
3. Create a directory structure
4. Implement storyboarding – prepare a sketch of contents, sequence, and layout of the GUI screen to be created
5. Build a prototype of your application
6. Test the prototype and implement the required changes
7. Deploy the application on the server

Before designing a Web-based application, you must first conceptualize the application. The goals and the objectives must be clearly defined. Think about what the Web-based application goals and missions may be – to Inform, Promote, Educate, Research and Report, or to simply Entertain. These goals must be defined and should be in line with the company's values and mission.

Annexure A shows a case study of a project.

1.3 Project Objectives

1. Develop an application based on the problem specifications given to you
2. Design a professional-looking design for the application
3. Integrate all the modules to form a complete solution

1.4 Project Deliverables

Following are the deliverables that have to be submitted on completion of the project:

1. Developed Web-based Application
2. User Manual
3. Installation Guide

1.5 Project Conduct

During the project development, three meetings would be held between the team.

- **First Meeting**



In this meeting, students need to discuss the eProject Specification. They need to work upon:

- Problem Statement
- Table/File Structure
- Screen/Forms to be created
- Program/Code Specifications
- Reports to be generated

- **Review of First Meeting**

After the first meeting, students will review and discuss the details of the First Meeting. They will check if the structure is correct and the Performance Statements are designed as per the design specifications.

➢ **Second Meeting**



In this meeting, students will discuss the design. They need to consider the following:

- Understanding of the problem
- Design of the application
- Validations required

➢ **Review of Second Meeting**

After the second meeting, students will review and discuss the details of the Second Meeting. They will check whether the design of the Web-based application is correct and proper verification and validation rules are being performed.

➢ **Third Meeting**

In this meeting, students will discuss and present the final Web-based application. Also consider the following:

- Implementing all the specifications mentioned in the previous meetings
- Form Design

- All the required validations
- Integrating all the project components such as the Menus, Forms, and Reports
- User Manual and Installation Guide



2. Steps for a Great Project

- Analyze and understand user requirements
- Do a detailed analysis for the project
- Identify the resources available
- Plan and maintain a schedule for the various activities to be done in the project
- Check and test all modules
- Document each and every detail

Documentation Section

In this section, we will look into various formats used for documenting the project.

This section contains different formats which can also be referred to, in Annexure A in the case study.

Students can use their own additional formats in the project documentation.

3. Documentation Section

This format can be used as the first page of your project report, duly signed by the Centre Academics Head.



This is to certify that

Mr./ Ms _____

Has successfully Designed & Developed

_____

Submitted by:

_____

Date of Issue:_____

Authorized Signature:_____

3.1 Problem Definition

Problem Definition:

In this form, you can define the problem statement. You can design a point wise list of the requirements as mentioned in the eProject Specification.

Form No.1/eProjects/Ver1.0

3.2 Customer Requirement Specification (CRS)

Client:

Business/Project Objective:

(Can address Organization/Business Overview, Products, Concerns, and Expectation from the System)

Inputs provided by the Client:

- Inputs for the Web-based application
- Outputs from the Web-based application
- Process Involved in the Web-based application
- Expected Delivery Dates
- List of Deliverables

Hardware Requirements:

Software Requirements:

Scope of the Work (in brief):

Form No.2/eProjects/CRS/Ver1.0

3.3 Architecture and Design of the Project

You can provide the Architecture and Design for your eProject in this format.
You can refer to Annexure A “Case Study” to understand the use of Architecture and Design of the Project.

Form No. 3/eProjects/Design/Ver1.0

3.4 Data Flow Diagram

A data flow diagram (DFD) is a graphic representation of the “flow” of data.

You can provide the Data Flow Diagrams for your eProject in this format.
You can refer to Annexure A “Case Study” to understand the use of Data Flow Diagrams.

Form No. 4/eProjects/DFD/Ver1.0

3.5 Flow Chart

A flow chart is a graphic representation of an algorithm or a process.

You can provide the flow charts for your eProject in this format.
You can refer to Annexure A “Case Study” to understand the use of flow charts.

Form No. 5/eProjects/FC/Ver1.0

3.6 Entity Relationship (ER) Diagram

Entity Relationship is used to describe information needs and/or the type of information that is to be stored in the database.

You can provide the Entity Relationship Diagrams for your eProject in this format.
You can refer to Annexure A “Case Study” to understand the use of Entity Relationship Diagrams.

Form No. 6/eProjects/ERD/Ver1.0

3.7 Database Design/Structure

Table is a set of data elements (cells) that is organized, defined and stored.

Using this format, you can provide the Table Structure for your eProject.
You can refer to Annexure A “Case Study” to understand the use of Table Structure.

Form No. 7/eProjects/TAB/Ver 1.0

3.8 Task Sheet

Project Ref. No.:		Project Title:	Activity Plan Prepared By:	Date of Preparation of Activity Plan:			
Sr. No.	Task			Actual Start Date	Actual Days	Team Member Names	Status

Form No.8/eProjects/TS/Ver1.0

3.9 Checklist of Validations

Option	Validated
Are all variables initialized with proper default values?	
Are the functions functioning according to their requirements?	
Is the output being displayed in the correct format?	
Is the output correct or accurate?	
Does the application's functionality satisfy the user requirements?	

Form No. 9/eProjects/VAL/Ver1.0

3.10 Submission Checklist

S. No.	Particulars	Yes	No	NA	Comments
1.	Are the displayed messages devoid of spelling mistakes?				
2.	Are the functions declared properly?				
3.	Have all the functions being tested?				
4.	Do the functions return correct values?				
5.	Are the different files integrated correctly?				

Form No. 10/eProjects/CKL/Ver1.0

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Annexure A

CASE STUDY: Online Auction System

Today the use of Internet has touched all the aspects of modern life. With the aid of the Internet, organizations are capable of doing business by reaching out to a global market to sell their products and services.

This helps number of people to buy and sell their products online by registering them on the e-commerce Websites. Apart from selling the products, the sellers are looking out for the best value and highest price for their products. Thus, many online Websites are providing the auction facility to their sellers.

'QuikBuynSell.com' is one of the e-commerce Website involved in buying and selling of the products through a Web-based online portal. The organization wants to offer their users with a facility of auctioning the products on the site. To accomplish the task, your team has been given the responsibility to develop an online auction system that will allow the users to negotiate over the buying and selling of goods over the Internet.

All users of the system must first register with the system. Once registered, they have to log on to the system for a particular session which will be valid for some time duration. Then, they will be able to sell, buy, browse the auctions, or bid the product placed in the auction on the Website.

A user who wishes to sell his/her product has to first register with the site. Once registered, the user who wishes to sell, will initiate an auction by informing the system of the items to auction, along with the minimum bid price, quantity of the products, reserve price for the goods, the start date of the auction, and the duration of the auction. For example, duration could be 15 days. The seller will have the right to cancel the auction as long as the auction has not started.

Customers who wish to follow an auction must first join the auction. Note, that it is only possible to join an active auction. Once a customer has joined the auction, he/she may make a bid, or post a message on the auction's bulletin board (visible to the seller and all customers who are currently participating in the auction). A bid is valid if it is over the minimum bid increment (to be decided by the system) and if the bidder has sufficient funds, that is, the customer's credit with the system is at least as high as the sum of all the pending bids. Bidders will be allowed to place their bids until the auction closes, and can place bids across as many auctions as they please. Once an auction closes, the system will first calculate whether the highest bid meets the reserve price given by the seller. If it is so, then the system will deposit the highest bid price minus the commission (to be determined by the system) taken for the auction service into the credit of the seller. The system will also notify the buyer and seller by sending e-mail with the details of the sale.

The auction system is highly concurrent -- customers will be bidding against each other in parallel, and a customer will place bids at different auctions and thus increase his/her credit in parallel.

ACKNOWLEDGEMENT

I would like to acknowledge all those who have given moral support and helped me to make the project a success.

I also express my sincere gratitude to the eProject Team at the Aptech Head Office and my project guide at the organization, for his valuable guidance and support for the completion of this project.

And finally, I would like to offer many thanks to all my colleagues for their valuable suggestions and constructive feedback.

Synopsis

The application to be developed for the online auction system will be a Web-based application. A seller or a bidder will logon to the site, and will be able to sell an auction item or bid an auction item. A bid will exist for a fixed duration, and after the fixed period the bid will expire. After successful bidding, the amount will be deposited to the seller's account and a small processing fee will be deducted. The system will have following functionality:

- Secure user registration
- Encrypted bid information
- Creating and viewing auction items
- Making a bid and validating a bid
- E-mail confirmation to buyers and sellers about successful or unsuccessful bidding
- Performing necessary bid value calculations after closure of an auction item, processing credit card related transactions and updating user accounts and maintaining transaction information
- Keeping track of transactions on all open and closed bids

The description and functionality of the different components of the Online Auction System is as follows:

- Web-tier

  The Web-tier contains a set of HTML pages, JSP pages and Servlet to register and validate the users of the site. This tier is responsible for authentication and authorization of users.

- Middle-tier

The Middle-tier contains a collection of EJB beans that implement business logic and persist the auction data that are generated during various operations in the database. A stateless session bean in this tier implements business logic, such as updating opening bid for an auction item, updating buyer and seller accounts after an auction closure, performing validation of a bid, besides doing usual site management chores, such as displaying an updated auction item list, or updating site auction history on successful bids.

When a user logs on to the site for the first time, the user profile is persisted into the database using the stateless session bean. The stateless bean persists the data using a registration entity bean.

When a seller first logs on to the site, and creates an auction item in the presentation tier successfully, the stateless session bean persists the auction item data in the database using an auction item entity bean.

When a buyer logs on to the site, and places a bid for an auction, it is first validated in the middle-tier, and encrypted with a symmetric key present in database. The encrypted bid is then entered in a JMS queue. The queue maintains a list of encrypted bids. An MDB picks one bid at a time from the queue, and then persists it in the database-tier.

When an auction closes, the Stateless Session bean retrieves all encrypted bids for the auction item by querying a bid entity bean. It then decrypts all bids using same symmetric key from database and persists the data. It finds the highest bid in the list, and then calculates the amount to be deposited in the seller's account and the amount to be deducted from the successful bidder's account. It then informs both buyer and seller, about the successful auction closure through e-mail, and updates the auction history database.

Another ID generator Stateless Session bean generates unique Item Id, Bid Id, and account number whenever a new item is added into the auction system.

A Web service provider in this tier, validates credit card details of both, the successful bidder, and the seller during an auction item closure. The Web service consumer is the stateless session bean.

Placing bid, retrieving a bid from the JMS queue, closing an auction item, and updating opening bid are transactional in nature.

The system implements a multi-threading transaction model to support concurrency. For every viewer, buyer, and seller, separate threads are created. Every buyer or seller transaction is maintained in respective threads. The system managed processes are carried out in a different thread.

- Database-Tier

A RDBMS fulfils the persistence requirements of the auction system. It maintains relational tables and facilitates data retrievals and modifications in an efficient manner. The auction system connects to database using JDBC connection pool.

Problem Definition

After reading the project specifications, the developer states the scope of the project very briefly. This is referred to as the problem definition. Queries related to the online auction system can be one or more of the following:

1. How will a seller enter an auction item?
2. How will multiple buyers bid for the same auction item?
3. How to validate a bid?
4. How can the bidding process be made secure?
5. How to implement approval of credit card transactions?
6. What kind of reports can be generated for the management?

Customer Requirement Specification (CRS)

Client: eMarket Place.

Business/Project Objective:
(Can address Organization/Business Overview, Products, Concerns, and Expectations from the System)

Create an online auction application where a seller or a buyer can log in and start auctioning process. The administrator can also log on to maintain user maintenance, auction items information, bid information, or generate reports for the management.

Inputs provided by the Client:

- Inputs for the Web-based application
- Outputs from the Web-based application
- Process involved in the Web-based application
- Expected Delivery Dates
- List of deliverables
- Data Constraints/Triggers/Validation Tables

Hardware and OS Requirements:

- Pentium IV CPU 2.80 GHz or Intel Core i3 or i5 processor
- 1 GB of RAM or higher
- Hard disk requirement: Minimum 4 GB
- Windows 7 or Windows 8

Software Requirements:

- NetBeans IDE 7.3 or higher
- Java EE 7 SDK
- SQL Server 2005
- Java SE 7.0
- Type 4 JDBC Driver for SQL Server 2012
- Internet Explorer/Google Chrome/Mozilla Firefox

Scope of the Work (in brief):

Depending on the decision given by the management, following are the technology requirements based on which the application needs to be developed:

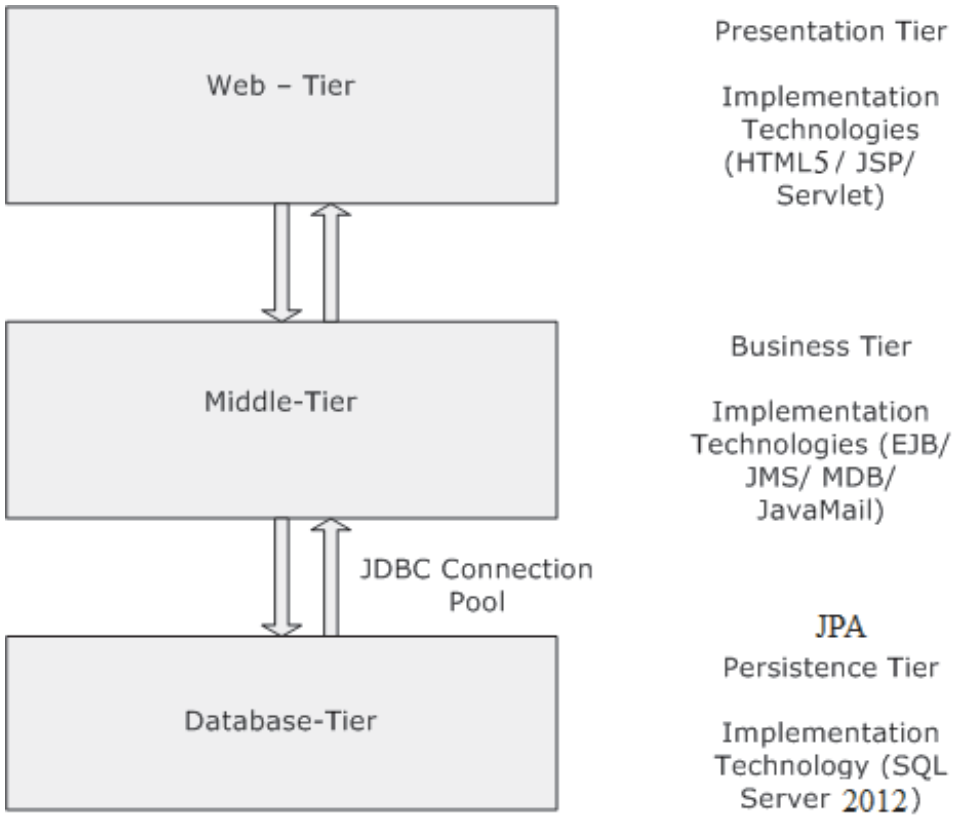
1. Storing application data in a relational database
2. Handling concurrency issues that will result from multiple buyers accessing the same auction item to place a bid
3. Managing multiple sessions for handling parallel bidding
4. Securing users and bid information
5. Handling credit card transactions

6. Handling transactions to maintain data integrity
7. Performing necessary calculations, and updating user accounts registered with the system
8. Notifying buyers and sellers after closing an auction

Form No.2/eProjects/CRS/Ver1.0

Architecture and Design of the Project

The application will be made of a Web-based distributed three-tier architecture to support multiple user transactions at the same time.

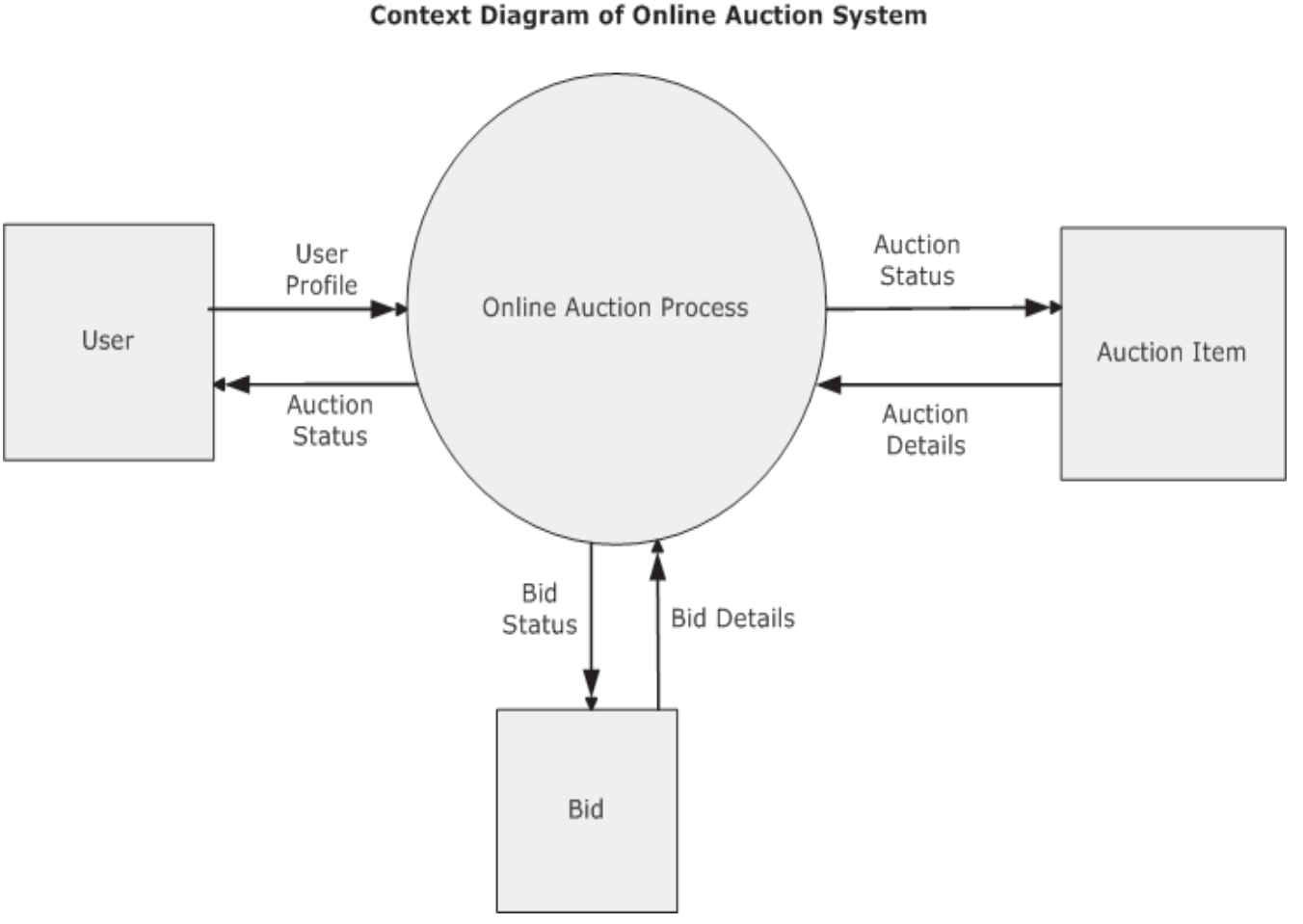


Web-Based Distributed Three-Tier Architecture of the Project

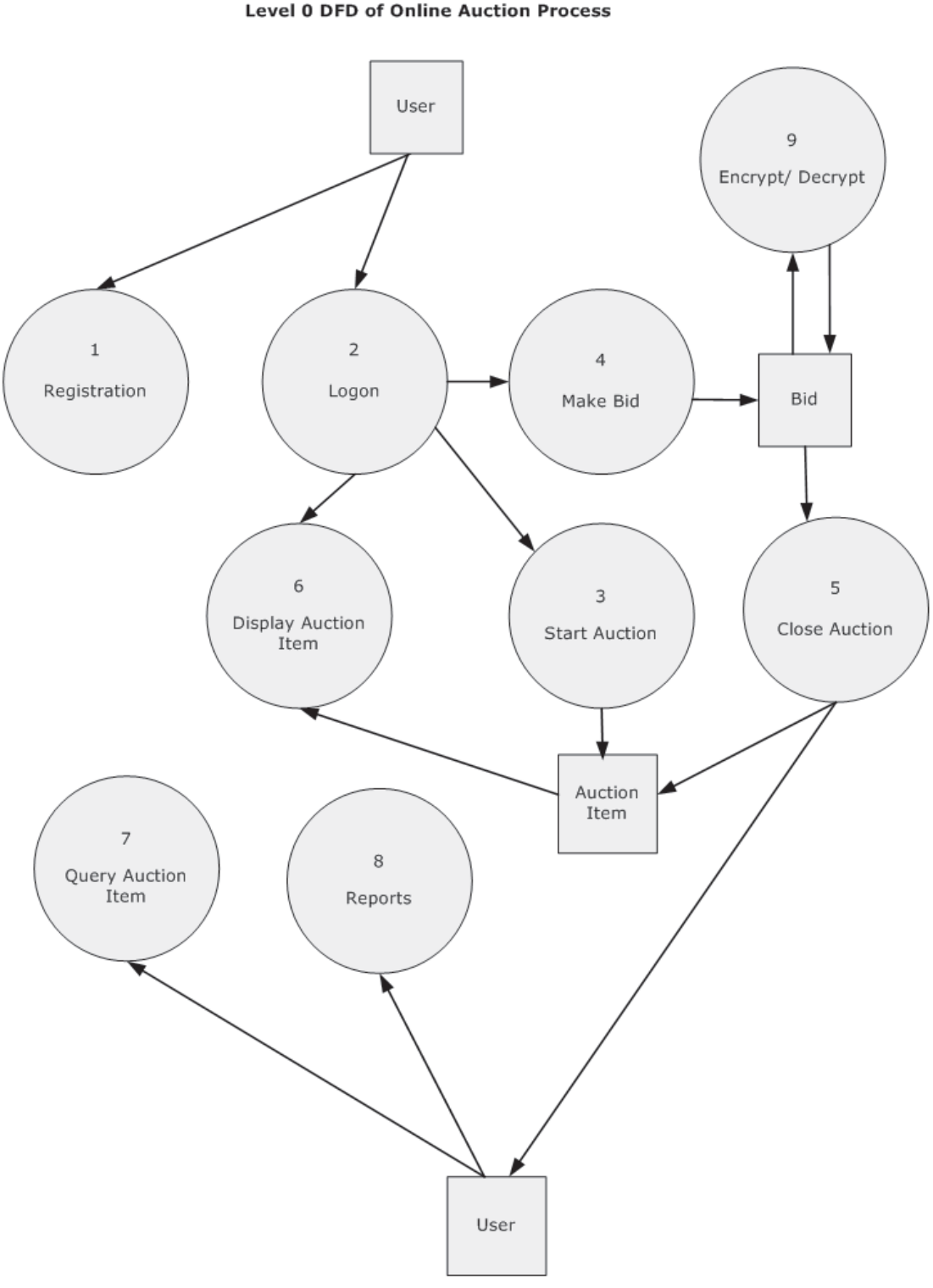
Form No.3/eProjects/Design/Ver1.0

Data Flow Diagram

The flow of data in the application is shown by the various data flow diagrams. The most basic data flow diagram is the Context diagram. It shows the basic flow of data in to and out of the system.

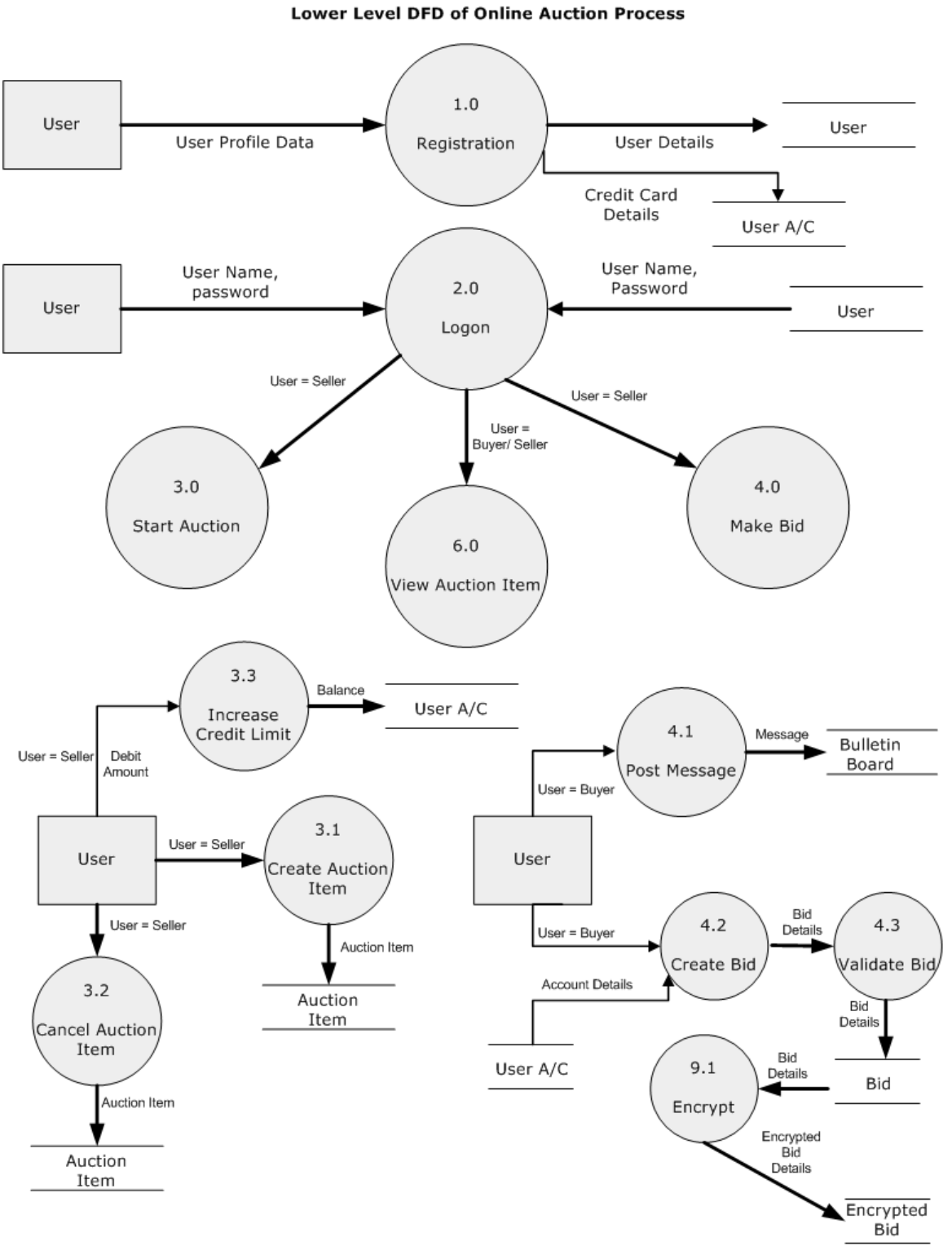


Context Diagram – Online Auction System



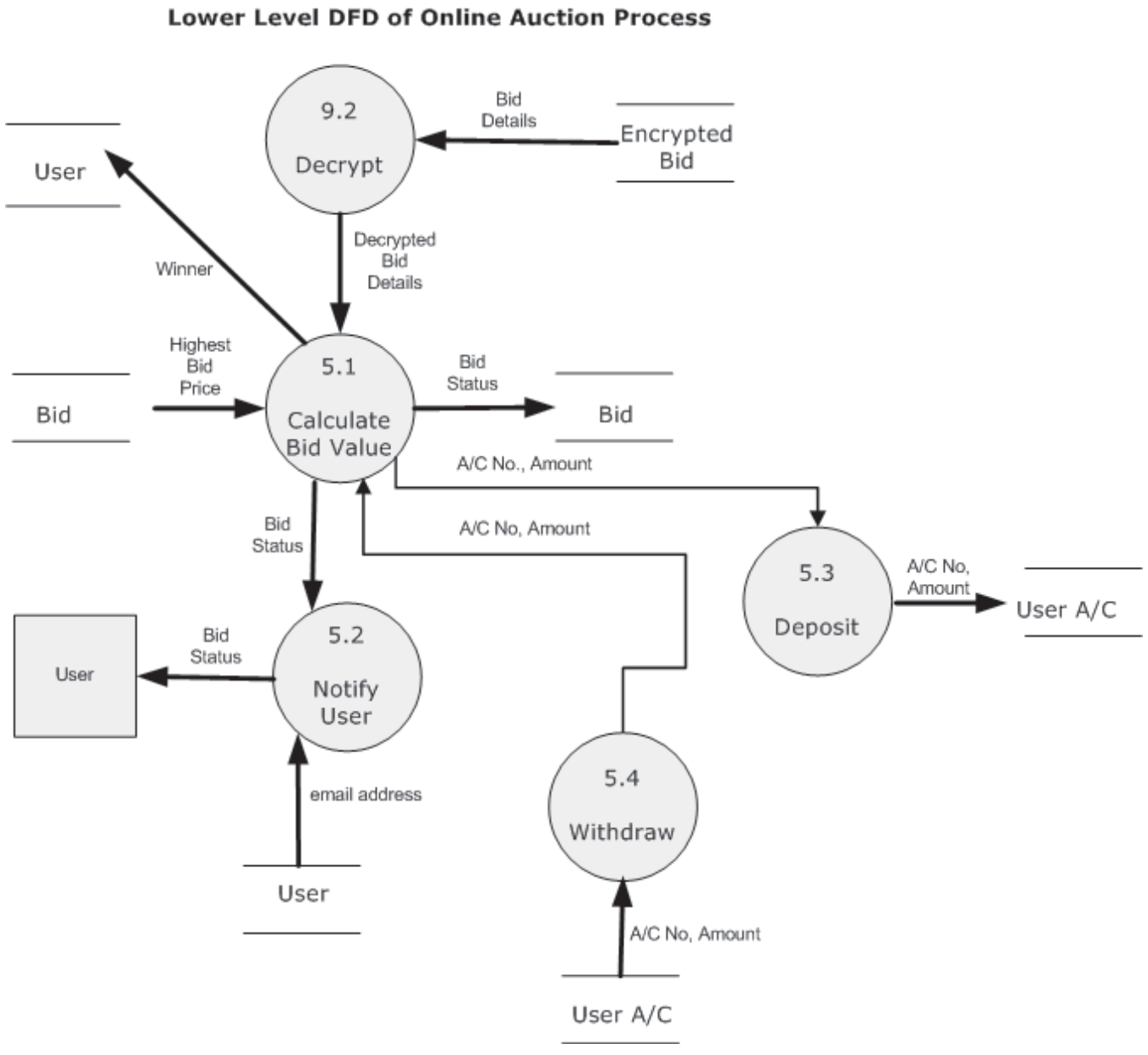
DFD LEVEL 0 – Online Auction Process

Form No. 4/eProjects/DFD/Ver1.0/2 of 4



Lower Level DFDs – Auction and Bidding Process

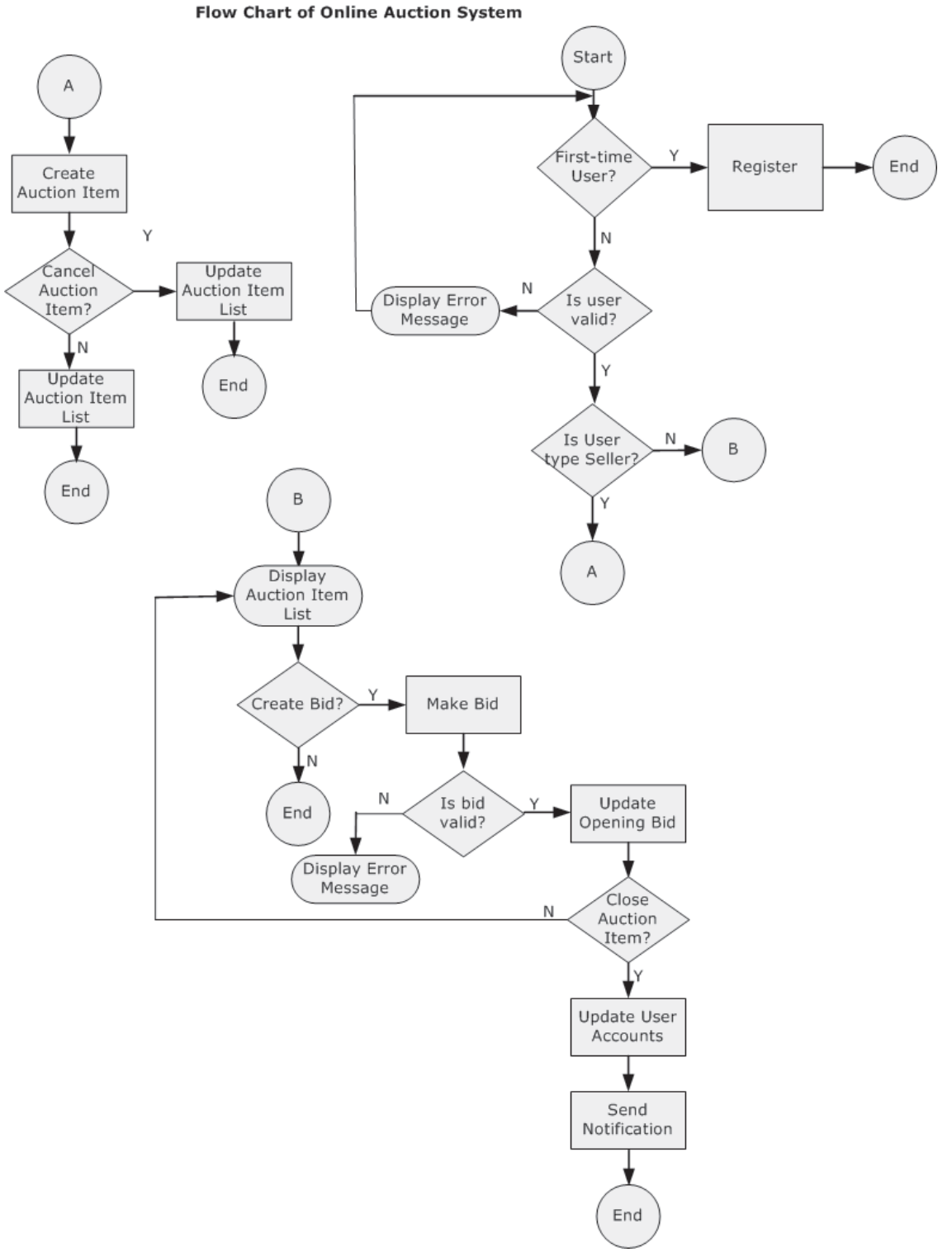
Form No. 4/eProjects/DFD/Ver1.0/3 of 4



Lower Level DFDs – Closing Process

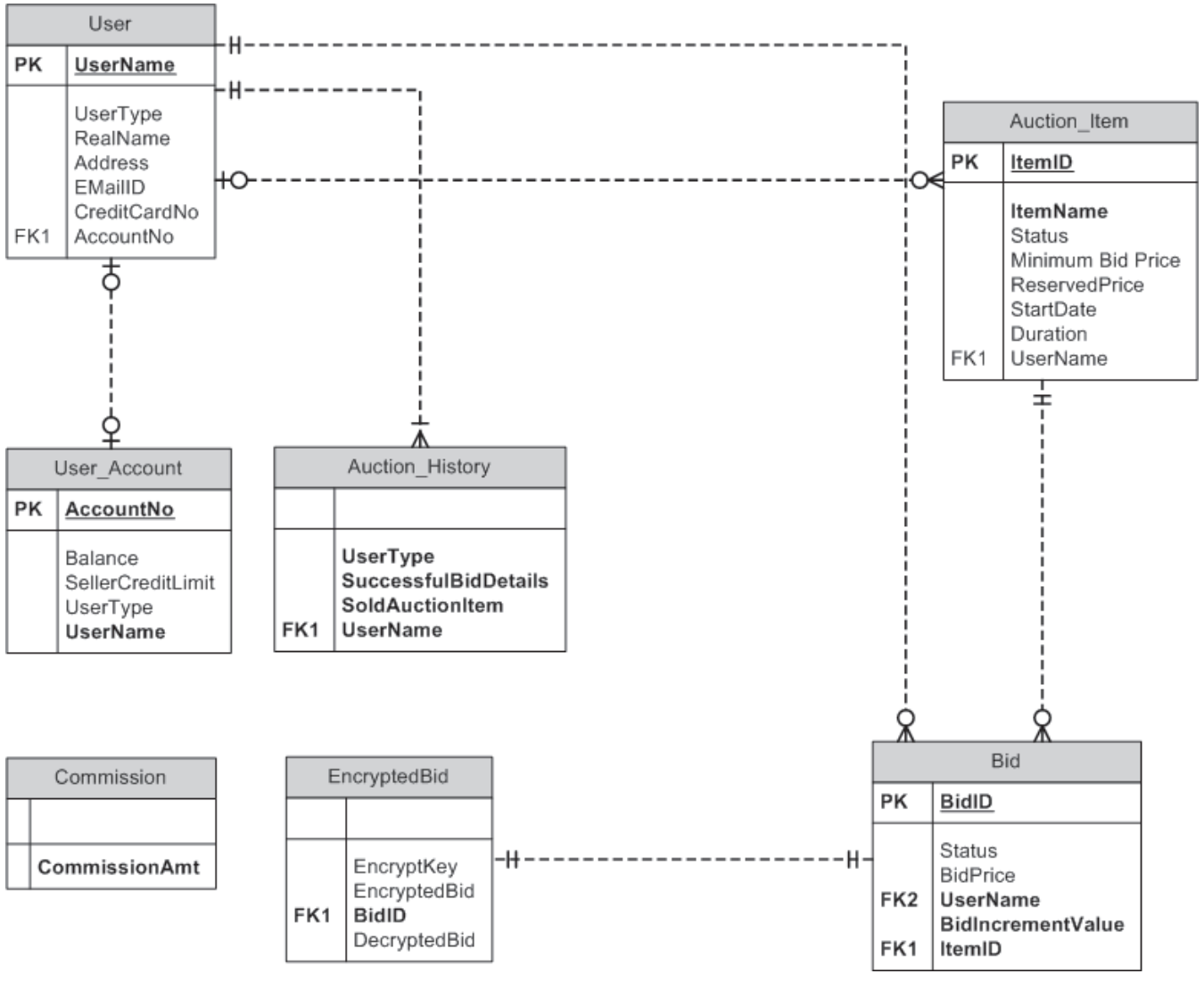
Form No. 4/eProjects/DFD/Ver1.0/4 of 4

Flow Chart



Form No. 5/eProjects/Flow Chart/Ver1.0

Entity Relationship (ER) Diagram



Form No. 6/eProjects/ERD/Ver1.0

Database Design/Structure

Table Design

Physical Name	Data Type	Req'd	PK	Notes
UserName	CHAR(20)	☑	☑	UserName identifies User
UserType	CHAR(10)	☐	☐	UserType is of User
RealName	CHAR(50)	☐	☐	RealName is of User
Address	CHAR(50)	☐	☐	Address is of User
EMailID	CHAR(20)	☐	☐	EMailID is of User
CreditCardNo	CHAR(20)	☐	☐	CreditCardNo is of User
AccountNo	CHAR(10)	☐	☐	AccountNo is of User

Table: User

Physical Name	Data Type	Req'd	PK	Notes
AccountNo	CHAR(10)	☑	☑	AccountNo identifies User_Account
Balance	NUMERIC(10,2)	☐	☐	Balance is of User_Account
SellerCreditLimit	NUMERIC(10,2)	☐	☐	SellerCreditLimit is of User_Account
UserType	CHAR(10)	☐	☐	UserType is of User_Account
UserName	CHAR(20)	☑	☐	UserName is of User_Account

Table: User_Account

Physical Name	Data Type	Req'd	PK	Notes
BidID	CHAR(10)	☑	☑	BidID identifies Bid
Status	CHAR(10)	☐	☐	Status is of Bid: Open/ Close
BidPrice	NUMERIC(10,2)	☐	☐	BidPrice is of Bid
UserName	CHAR(20)	☑	☐	UserName is of BidderName
BidIncrementV...	NUMERIC(10,2)	☑	☐	BidIncrementValue is of Bid
ItemID	CHAR(10)	☑	☐	ItemID is of Bid

Table: Bid

Physical Name	Data Type	Req'd	PK	Notes
ItemName	CHAR(50)	☑	☐	ItemName identifies Auction_Item
Status	CHAR(10)	☐	☐	Status is of Auction_Item: Open/ Closed/Cancelled
Minimum Bid Price	NUMERIC(10,2)	☐	☐	Minimum Bid Price is of Auction_Item
ReservedPrice	NUMERIC(10,2)	☐	☐	ReservedPrice is of Auction_Item
StartDate	DATETIME	☐	☐	StartDate is of Auction_Item
Duration	DATETIME	☐	☐	Duration is of Auction_Item
ItemID	CHAR(10)	☑	☑	ItemID identifies Auction_Item
UserName	CHAR(20)	☐	☐	UserName is of Auction_Item

Table: Auction_Item

Physical Name	Data Type	Req'd	PK	Notes
UserType	CHAR(10)	☑	☐	UserType is of Auction_History: Buyer/ Seller
SuccessfulBidDetails	VARCHAR(100)	☑	☐	SuccessfulBidDetails is of Auction_History
SoldAuctionItem	VARCHAR(100)	☑	☐	SoldAuctionItem is of Auction_History
UserName	CHAR(20)	☑	☐	UserName is of Auction_History

Table: Auction History

Form No. 7/eProjects/TAB/Ver1.0/3 of 3

Task Sheet

Project Ref. No.: eP/MUSICWORLD/01		Project Title:	Activity Plan Prepared By:	Date of Preparation of Activity Plan:			
Sr.No.	**Task**			**Actual Start Date**	**Actual Days**	**Team Member Names**	**Status**
1	GUI Design	Online Auction System	John Doe, Quick BuyNSell Services	03-01-15	5	Kate	Completed

Form No.8/eProjects/TS/Ver1.0

Checklist of Validations

Option	Validated
Are all the Input masks consistent?	Yes
Do all numeric fields have a default value of 0?	Yes
Can a user register with the application after validation is performed on the User ID and password provided by the bidder?	Yes
Does the seller have the option to log in?	Yes
Can the seller create and delete all the records regarding the items auctioned?	Yes
Does the auction item get updated once a customer purchases an item?	Yes
Does each unit meet its objective and purpose?	Yes
Are all the validations working as specified in Process Design?	Yes
Does the code work as per specifications?	Yes
Does the application's functionality resolve the client problem, and satisfy his needs completely?	Yes

Form No. 9/eProjects/VAL/Ver1.0

Submission Checklist

Sr. No.	Particulars	Yes	No	NA	Comments
1.	Are the users able to register with the application after validation is performed on the UserID and password?	Yes			
2.	Are the customers able to search for a particular auction item?	Yes			
3.	Are the GUI contents devoid of spelling mistakes?	Yes			
4.	Is the administrator able to generate the required reports?	Yes			
5.	Are all the GUI components working?	Yes			Tested on Windows 7 and Windows 8
6.	Is the application user friendly?	Yes			

Form No. 10/eProjects/CKL/Ver1.0

A. **System Requirements:**

No.	Items	Description
1.	Operating System	Windows 7 or Windows 8
2.	Database	MS SQL Server 2012
3.	Software	JDK 1.7, NetBeans 7.3 or higher, Type 4 JDBC Driver for MS SQL Server 2012, and Java EE 7 SDK.

B. **Install and Run Application:**

Step 1: Copy the **QuickNSell** Software Service folder in the appropriate directory.

Step 2: Execute the application.

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